

The North Carolina School of Science and Mathematics

Polyphonic Piano & Sequencer

Gabriel Kinney

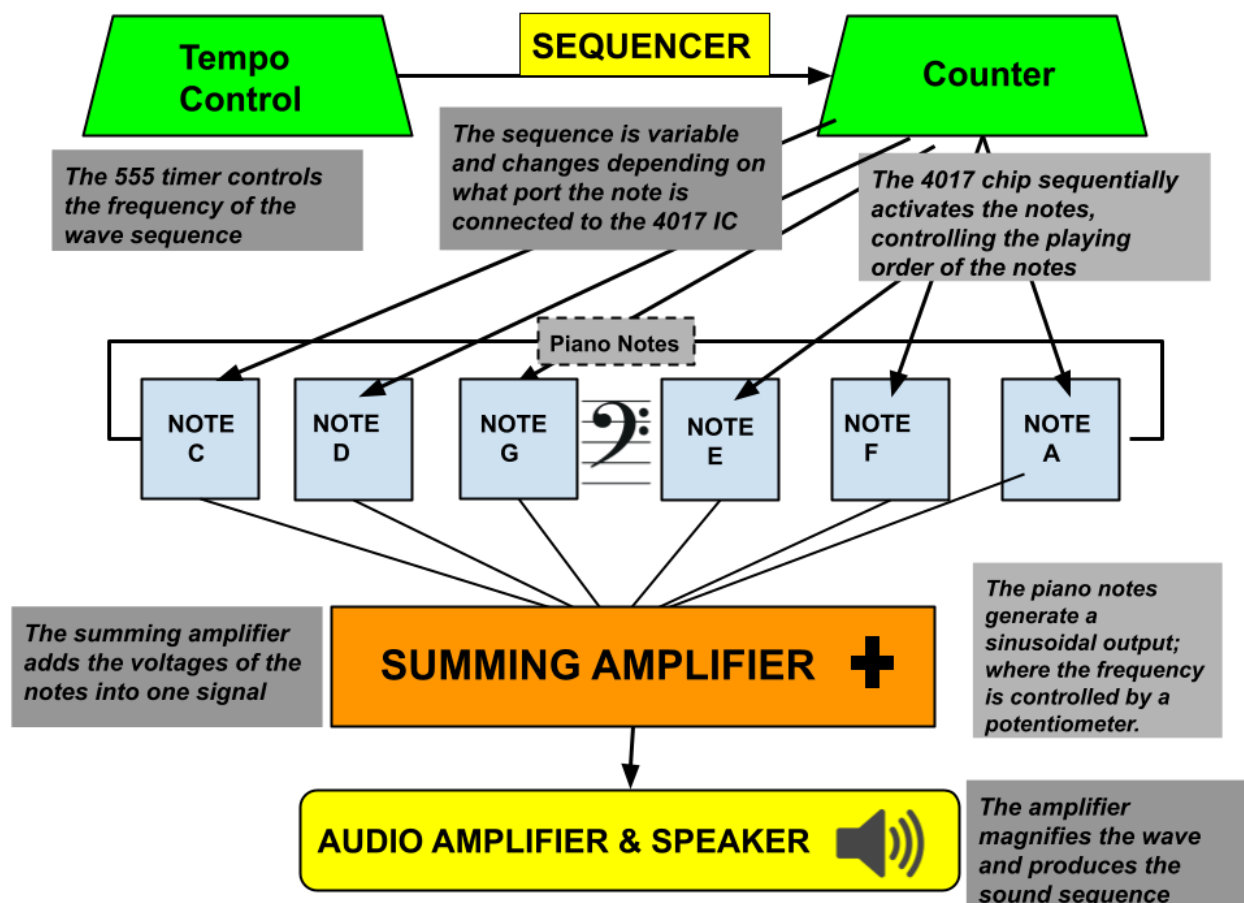
EE 4020: Electrical Engineering

Mr. Kirk

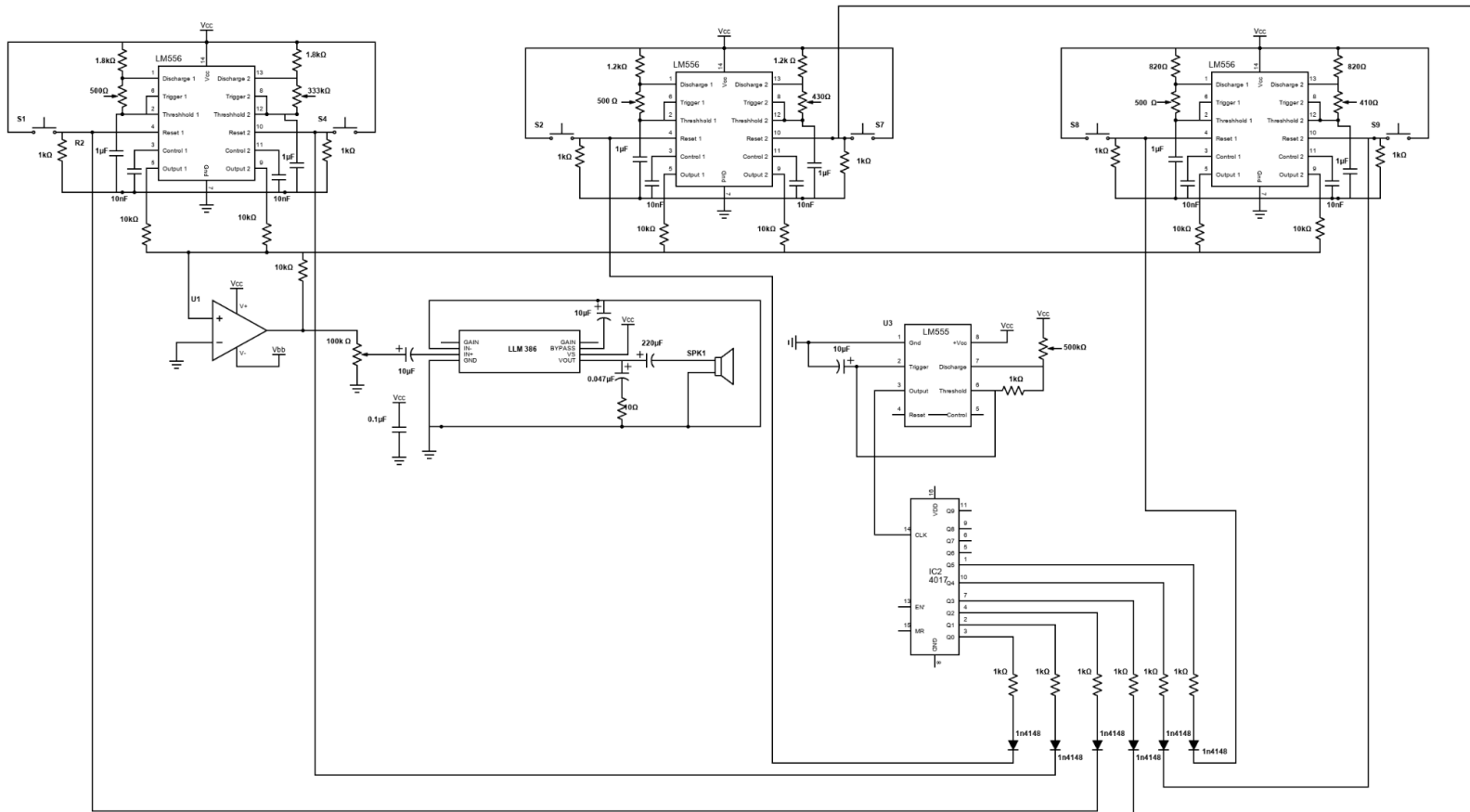
December 10, 2025

How does this work?

The Polyphonic Piano works uses a series of 556 timers in an astable multivibrator circuit to produce the desired frequency that matches the note range. There are 6 notes in the C5 octave on a standard piano. The specific frequency is set by a 500k Ω potentiometer that adjusts the waveform's frequency. All 6 notes are fed into a LM741 summing amplifier circuit that combines their voltages (signals) into a single signal. Once the signal is produced, it is directed to an LM386 audio amplifier and speaker circuit, which amplifies the waveform and converts the signal into sound. The sequencer is composed of an LM555 timer and a 4017 Integrated Circuit. The LM555 controls the frequency of the final waveform, which determines how often the notes play. Lastly, the 4017 Integrated Circuit sequences the notes in the desired order. Finally, producing a sequence that creates a melody.



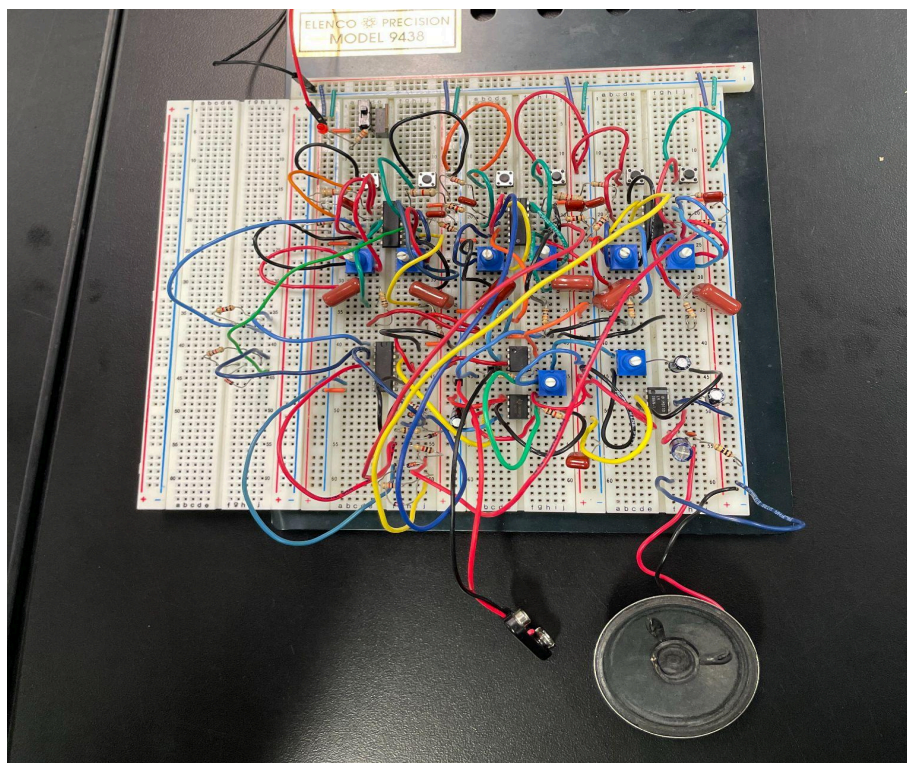
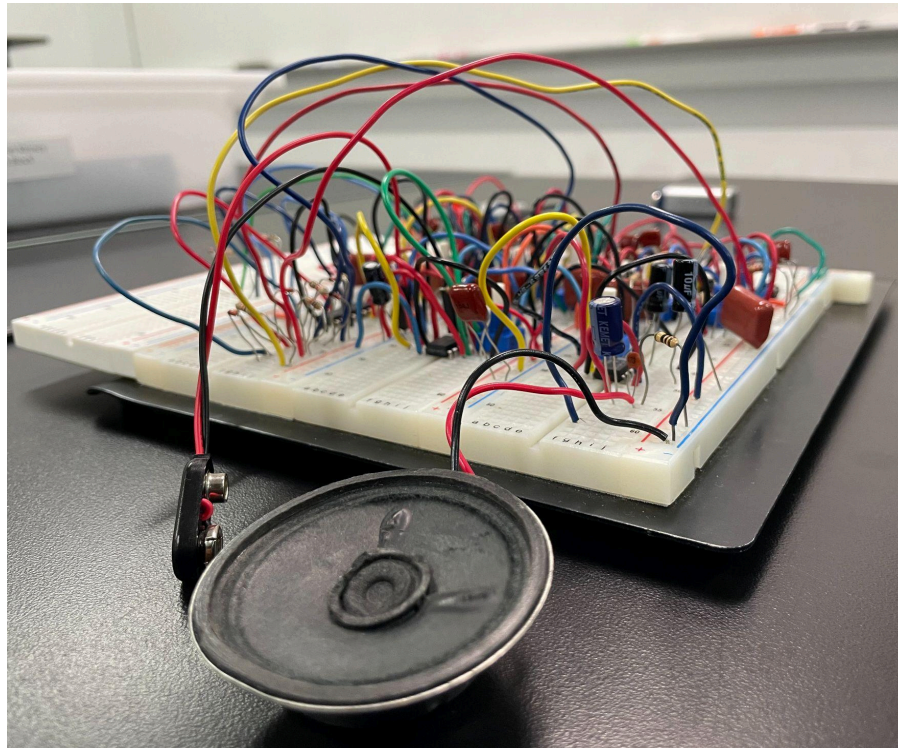
Schematic



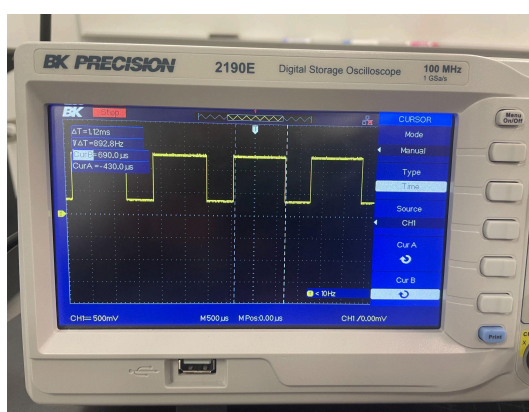
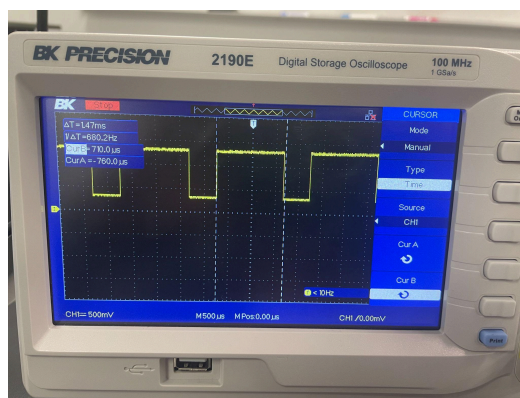
How is this different from an electric piano?

The digital polyphonic piano is vastly different than an electronic piano. Electric pianos use microprocessors to reproduce realistic acoustic piano tones. This system produces musical notes solely through square-wave oscillation. Additionally, while commercial electric pianos achieve high polyphony through digital signal processing, this circuit is limited by the number of oscillators and physical components available. Compared to a true electric piano, which generates sound through mechanical and electromagnetic parts, this breadboard piano contains no moving sound-producing parts.

Circuit Photos



Waveform Photos



Parts List

PART TYPE	Component Description	Quantity	Notes
Integrated Circuits (ICs)	556 Timer	6	Dual timer IC
	555 Timer	1	Single timer IC
	4017 Chip	1	Decade counter/divider (CMOS)
	LM386	1	Low-voltage audio power amplifier IC
	LM741	1	Operational amplifier (Op-amp) IC
Resistors	1 k Ω Resistor	13	
	1.8 k Ω Resistor	2	
	1.2 k Ω Resistor	2	
	820 Ω Resistor	2	
	10 k Ω Resistor	7	
	10 Ω Resistor	1	
Potentiometers	500 Ω Potentiometer	5	Variable resistor
	1 k Ω Potentiometer	1	Variable resistor
	100 k Ω Potentiometer	1	Variable resistor
	500 k Ω Potentiometer	1	Variable resistor
Capacitors	1 μ F Capacitor	7	Microfarad
	10 nF Capacitor	6	Nanofarad
	10 μ F Capacitor	2	Microfarad
	0.1 μ F Capacitor	1	Microfarad
	0.047 μ F Capacitor	1	Microfarad
Semiconductors	1N4148 Diode	6	High-speed switching diode
Output/Miscellaneous	Speaker	1	
	9v Batter	2x	

Reflections

How did this evolve from the initial design during the building process?

- Initially, the Polyphonic Piano was a fundamental circuit with 6 LM556 timers that produced different waveforms. Soon after, it was a digital piano that was able to produce 6 in-tune notes. It finally evolved into a fully functioning circuit that summed the signals, amplified the waveform, adjusted tempo, controlled sequence, and produced audio.

What went well, what went wrong, and why?

- I did well with my calculations for my unknown part values, which included capacitors, resistors, and potentiometers. However, I struggled with wiring as I used wires that were longer than needed and did not plan the orientation of my circuit on the breadboard. This made it difficult to troubleshoot my circuit and also to install later sections of the circuit. I fixed this by doing some rewiring throughout the build to make things more organized, and I also added an extension to the far left side to provide room to expand. I also had difficulty with my parts list in the end, as I did not create a table before building, but after my circuit was complete. This made it difficult to get a components list in the end, as I had to remove parts to figure out their values. These revisions were not instantaneous. It took well over 5 hours of troubleshooting to implement the fixes, which yielded the current build.

What would you do differently next time?

- Next time I work on an electronic circuit, I would like to tidy up my wiring, take better notes of the calculated values, and keep track of my parts early on throughout the build. Overall, I would like to focus more on the organization of my circuit building in the future to streamline the process and cut down on the time it takes. If more time had been spent on the organization, the build would have taken less time, as there would have been no need to backtrack.